

LT 8130 Sociocultural and Design-Based Foundations

Final Project

Vimala Telu

Introduction

MagicSchool AI is among the most widely used AI tools in education today, with over five million teachers registered across the United States and beyond. When I first explored the platform, I was genuinely impressed by how clean, fast, and well-designed it is as a time-saving tool. But the more I used it, the more a different question took shape: what kind of learning does this tool actually support, and what kind does it make impossible? As AI increasingly mediates educational experiences, this question feels urgent not as a critique of technology itself, but as a question about how AI-enhanced tools can better preserve what sociocultural theory tells us matters most for genuine learning.

This project examines MagicSchool AI through the sociocultural and design-based frameworks. I describe the tool and the assumptions about learning embedded in its design. I analyze where it succeeds and where it falls critically short of what participation in communities of practice, identity development, cultural context, and human connection require. I propose a concrete redesign — grounded in Lave and Wenger’s communities of practice, Gutiérrez and Jurow’s equity-by-design, Sfard’s participation metaphor, and Moll et al.’s funds of knowledge — that transforms MagicSchool AI from a solo content-production tool into a genuine community-based professional learning platform.

Part 1: Describing MagicSchool AI

1.1 Background, Audience, and Learning Goals

MagicSchool AI was created in 2023 by Adeel Khan, a former classroom teacher, to provide educators with AI tools designed for their work. It is free, requires only a Google account, and

has more than five million registered teachers worldwide. The platform offers more than 80 tools for lesson planning, differentiation, assessment, rubric design, family communication, and professional support. Its stated learning goals are practical: improve the quality of instructional materials teachers produce and reduce the time spent on planning and documentation so teachers can focus more on students. The platform measures its success by hours saved per week, a metric that reflects exactly the kind of learning it imagines.

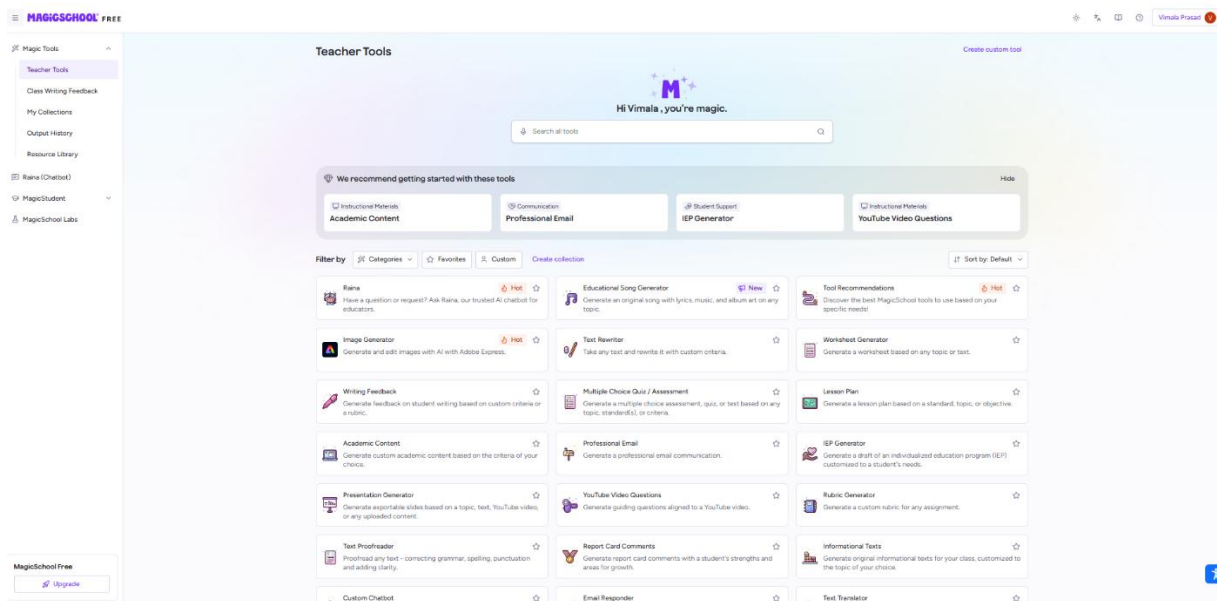


Figure 1: MagicSchool AI dashboard showing the grid of over 80 individual teacher tools.

1.2 How AI Functions and the Role It Assigns to Learners

The AI functions as an expert content producer. A teacher opens the tool, fills out a form, and receives a complete document within seconds. The Lesson Plan Generator — which I explored in detail — has four fields: grade level, topic or standard, an optional Additional Criteria field with no guidance on student context, and a standards alignment field. I entered Grade 5 and the fractions standard CCSS.MATH.CONTENT.5.NF.A.1, left Additional Criteria blank (as most teachers likely do), and clicked Generate. A full lesson plan appeared in about ten seconds. The

teacher's role is entirely reactive: specify parameters, receive output, and adapt. There is no moment when the tool asks who the students are, what the teacher has tried before, or what professional question is driving the request.

The screenshot displays the 'Prompt Editor' interface for the MagicSchool AI Lesson Plan Generator. It features four distinct input fields for generating a lesson plan:

- Lesson Plan:** A header section with a 'Show exemplar' link. Below it, a description states: 'Generate a lesson plan based on a standard, topic, or objective.'
- Grade level: ***: A dropdown menu currently set to '5th grade'.
- Topic, Standard, or Objective: ***: A text area containing the prompt: 'Fractions — Adding and Subtracting Fractions with Unlike Denominators (CCSS.MATH.CONTENT.5.NF.A.1). Students will use equivalent fractions to add and subtract fractions with unlike denominators, including mixed numbers, and solve word problems involving addition and subtraction of fractions with unlike denominators.'
- Additional Criteria:**: A text area containing the prompt: 'Students are in a unit about world regions, students last lesson was on the geography of the United States, have the lesson include group work, etc.'
- Standards Set to Align to:**: A text area containing the prompt: 'CCSS — Common Core State Standards (Mathematics, Grade 5)'

Each text area includes an 'Add File' button and a 'Total word limit: 0/75,000' indicator. A 'Generate' button is located at the bottom, along with a 'Web' icon.

Figure 2 MagicSchool AI Lesson Plan Generator showing four input fields.

1.3 Learning Assumptions Embedded in the Design

MagicSchool AI is built on what Sfard (1998) calls the acquisition metaphor — knowledge is packaged and transferred to an individual teacher, and learning is measured by the efficiency of that transfer. What is entirely absent is the participation metaphor: the understanding that professional learning happens through becoming part of a community, engaging in shared inquiry, and developing expertise alongside colleagues over time. The platform treats teacher

development as a solo, transactional activity — a design choice with significant consequences for the kind of professional learning it enables.

Part 2: Sociocultural and Design-Based Analysis

Participation in Communities of Practice

Lave and Wenger (1991) argue that people become more capable by participating alongside others in shared practices — not by receiving information alone. For teachers, this means working through real challenges with colleagues, observing each other’s practice, and building shared professional knowledge. MagicSchool AI has no mechanism for this. I searched the platform for collaborative features using the terms ‘collaborate,’ ‘team,’ and ‘co-plan.’ What I found was a prompt-sharing library, a place to exchange AI inputs, not a community of professional inquiries. The screenshots below show the full navigation and search results, highlighting the absence of community infrastructure.

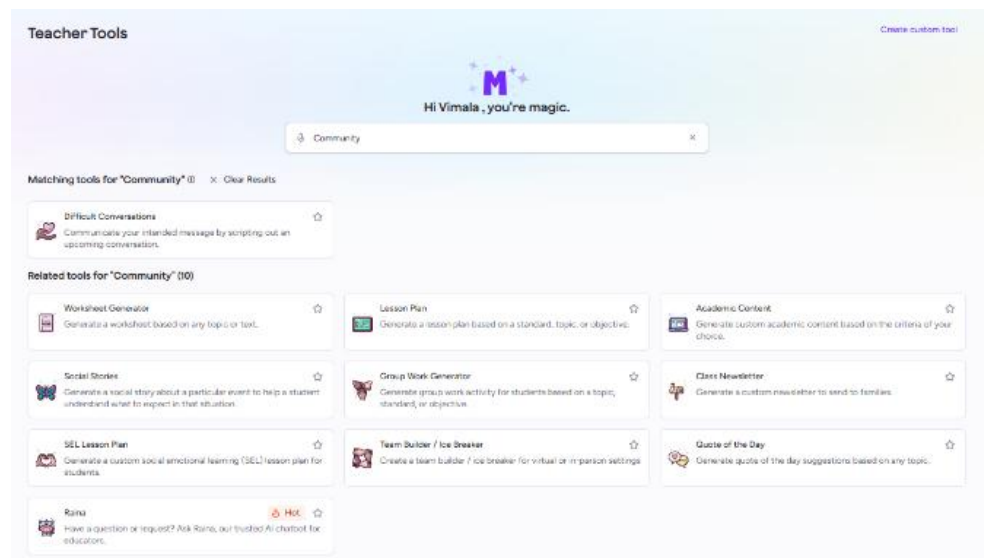


Figure 4: MagicSchool AI navigation showing all available platform sections

I also searched for collaborative tools directly, using the search terms ‘collaborate,’ ‘team,’ and ‘co-plan.’ The results either returned nothing relevant or tools that, on closer inspection, were

still designed for a single teacher working alone. The screenshot below shows what that search produced.

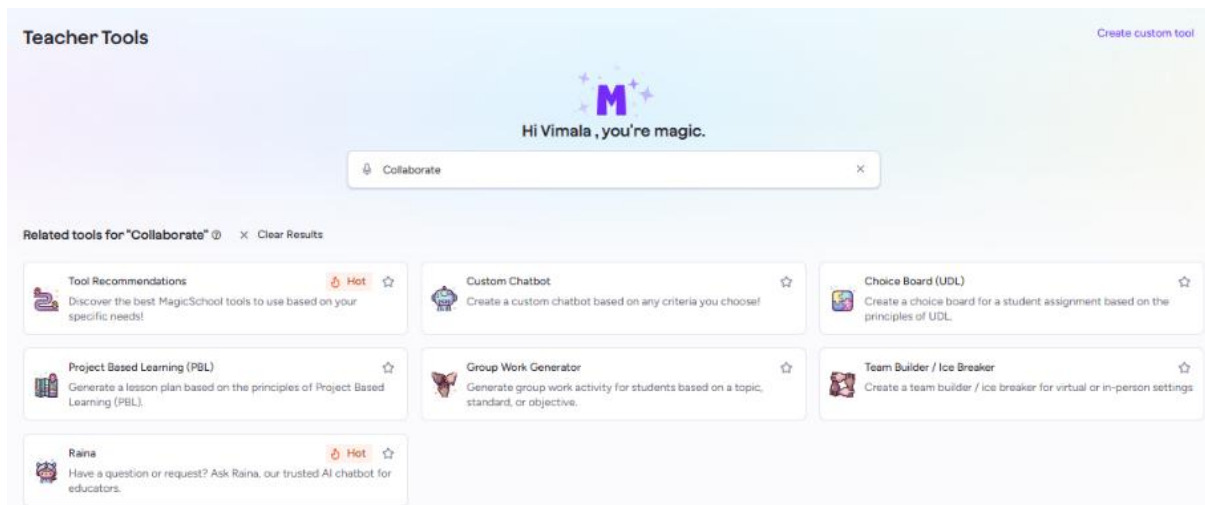


Figure 5: Search results in MagicSchool AI for collaborative and community-oriented features

Identity Formation, Funds of Knowledge, and Situated Context

Professional identity develops through participation in a community of practice — through being recognized as a practitioner who notices student thinking, builds on community knowledge, and learns alongside colleagues. MagicSchool AI consistently positions teachers as efficient individual content producers and offers no tool that invites reflection on student thinking or collegial professional inquiry. Regarding funds of knowledge, Moll et al. (1992) showed that students carry rich cultural and community knowledge that should be central to instructional design. Yet every form I encountered in MagicSchool AI asked only about academic parameters. I checked the differentiation tools specifically.

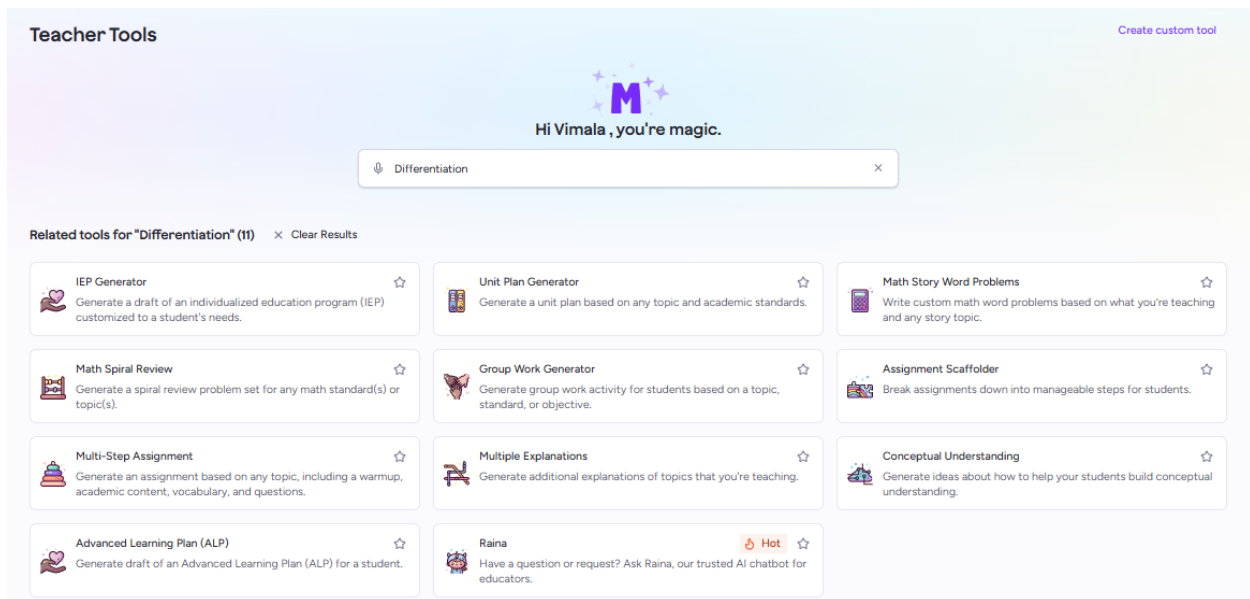


Figure 6 MagicSchool AI differentiation tool input form, showing that differentiation

A lesson generated for a bilingual Latinx classroom in Atlanta is identical to one generated for any other setting because community knowledge is never incorporated into the design. This is the opposite of what Gutiérrez and Jurow (2016) mean by equity-by-design. Lave and Wenger (1991) note that learning is always situated — inseparable from the specific context in which it occurs. MagicSchool AI’s outputs are thoroughly decontextualized: no school, no community, no real students. The most important pedagogical work, making generic content meaningful in a specific context, happens entirely outside the tool, with no support.

Embodiment, Power Relations, and Unrealized Potential

Embodied cognition learning is not only cognitive but also physical and relational. MagicSchool AI produces only text-based documents and offers no tools to observe or support the embodied dimensions of classroom learning. On power, Gutiérrez and Jurow (2016) argue that design is never neutral: when an AI determines what a good lesson looks like without ever asking who the students are, it encodes dominant assumptions about whose knowledge counts. Teachers who

adopt generated materials uncritically may reproduce those assumptions without realizing it. Despite these significant limitations, MagicSchool AI has genuine unrealized potential. It reaches five million teachers — a scale most professional development programs never approach — and its existing technical infrastructure could be redirected to facilitate shared inquiry rather than individual content production. That potential is the foundation of the redesign below.

Part 3: Redesigning MagicSchool AI

3.1 Overview and Theoretical Grounding

My redesign works at two levels: a concrete feature change to the existing Lesson Plan Generator (the Community Context Intake), and a new platform module (Asynchronous Lesson Study Groups). Together they shift MagicSchool AI from the acquisition to the participation metaphor, from individual productivity to community-based professional learning, and from generic to situated, community-informed instruction. Every design decision is anchored in course frameworks: Lave and Wenger's (1991) communities of practice provides the structural architecture; Gutiérrez and Jurow's (2016) equity-by-design shapes the intake; Sfard's (1998) participation metaphor reorients the measure of success; Moll et al.'s (1992) funds of knowledge shapes what the intake asks; Gutiérrez's (2008) third space shapes the bridge between official curriculum and community knowledge; Week 8 embodiment informs the activities generated; and the Week 4 identity framework informs how the redesign addresses professional identity development the current platform ignores entirely.

3.2 Feature 1: Community Context Intake

Before the current four-field form appears, teachers answer eight questions — one for each sociocultural dimension. Question 1 asks who the students are, including home languages and

cultural backgrounds, activating funds of knowledge from the start (Moll et al., 1992). Question 2 asks how students physically experience this concept outside school, surfacing embodied knowledge (Week 8). Question 3 asks what is happening in the classroom and school right now, grounding design in situated context (Lave & Wenger, 1991). Question 4 asks how students see themselves in this subject, naming the identity gap between lived competence and school identity (Week 4). Question 5 asks whose knowledge the standard curriculum excludes, making power relations explicit (Gutiérrez & Jurow, 2016). Question 6 asks which students this lesson might leave behind and why, embedding equity-by-design. Question 7 asks what bridge exists between what students already know and what is being taught, creating the third space (Gutiérrez, 2008). Question 8 asks what professional question the teacher is exploring with a colleague, connecting individual planning to collective inquiry. Each answer shapes the AI's output. A teacher who provides full community context receives a fundamentally different lesson than one who only fills in grade and topic.

M
magicschool.ai

My Tools
Lesson Study Groups
Knowledge Base

VT

Step 1 of 2 — Community & Student Context

SCREEN 02 — REDESIGN: Community Context Intake — 8 Questions (NEW feature)

REDESIGN: Before any lesson is generated, teachers answer 8 questions — one for each sociocultural dimension. These answers transform every output from generic to community-grounded.

Tell us about your students and community.

Your answers shape every example, cultural connection, and observation prompt in the lesson we generate for you. Takes about 2 minutes.

1

Who are your students? *

Home languages, cultural backgrounds, family context, community.

My students are 5th graders at a Title I school in Atlanta, GA. Most come from Latinx families where Spanish is spoken at home. Parents work in food service, construction, and agriculture. Students have very strong intuitive math skills from helping at home and in family businesses.

5

Whose knowledge is centered in how this topic is usually taught? *

Whose examples, methods, ways of knowing appear in the curriculum?

Standard instruction uses textbook examples — pizza diagrams, abstract number lines, formal symbols. My students' ways of knowing fractions — through cooking, dividing, measuring in real life — are never the starting point. Curriculum centers one kind of mathematical knowing.

2

How do students physically experience this concept in daily life? *

What they touch, make, measure, move — at home, in community.

Students divide food daily — pan dulce, tortillas, fruit — into equal parts for family meals. They measure cooking ingredients and several help parents measure fabric for sewing or calculate supplies for construction jobs. Fractions are embodied daily, just never named formally.

6

Which students might this lesson currently leave behind, and why? *

Consider language, prior experience, structural barriers, access.

ELL students struggling with math vocabulary in English. Students who have internalized 'bad at math' identity. Students managing heavy responsibilities outside school. Students whose family's mathematical practices differ from textbook format.

3

What is happening in your classroom/school right now? *

Current unit, recent events, assessment context, school dynamics.

3 weeks into fractions unit. Students passed conceptual portion but struggle with symbolic notation. Standardized tests in 6 weeks. High ELL population. Many students freeze when they see the p/q format even though they understand the concept.

7

What is the bridge between what students already know and what you are teaching? *

Name the specific connection — where existing knowledge meets the new concept.

My students already know how to divide food and objects fairly into equal parts — they understand fractions physically and intuitively. The bridge: 'when you cut the tortilla into 4 equal pieces, each piece is one-fourth' — then show p/q is just how we write what they already know.

4

How do students see themselves in relation to this subject? *

Do they feel capable, excluded, anxious? What shapes their math identity?

Many say they are 'not math people' — but solve complex proportion problems mentally when dividing ingredients or calculating prices at family stores. Deep disconnect between their lived competence and their school identity as math learners.

8

What professional question are you exploring with a colleague? (optional)

Share your shared inquiry question if part of a lesson study group.

My colleague Ms. Johnson and I are exploring: how do we bridge students' informal fraction knowledge from cooking and community life into formal symbolic notation? Please embed an observation prompt we can both use to compare what we see across our classrooms.

Next: Add lesson details →

Figure 7. Prototype showing the redesigned Community Context Intake with eight questions.

A teacher who describes Latinx bilingual students with embodied fraction knowledge through daily cooking receives a lesson opening with pan dulce and tortilla-folding examples, key vocabulary in English and Spanish, a physical measuring activity, and an observation prompt connected to her shared inquiry with a colleague — each change following directly from the community context she provided.

Step 2 of 2 — Lesson Details

SCREEN 03 — REDESIGN: Community-Informed Lesson Generation [Step 2 of 2]

Community context saved: Latinx students · bilingual Spanish-English · cooking & agriculture · peer inquiry with Ms. Johnson on fraction bridging | This shapes your entire lesson.

Now, tell us about the lesson.

Your community context from Step 1 will shape the examples, language support, embodied activities, and observation prompts we generate.

Grade level: *
5th grade ▼

Topic, Standard, or Objective: *
Fractions — Adding and Subtracting Fractions with Unlike Denominators (CCSS.MATH.CONTENT.5.NF.A.1)
Students will use equivalent fractions as a strategy to add and subtract fractions with unlike denominators, including mixed numbers.

Standards Set to Align to:
CCSS — Common Core State Standards, Mathematics Grade 5

Preview — your output will include:

- Cultural anchor:** Lesson opens with pan dulce and tamale-making — dividing food into equal parts — connecting to unlike denominators
- Bilingual support:** Key vocabulary presented in English and Spanish: 'denominator/denominador,' 'equivalent/equivalente'
- Embodied activity:** Students physically fold and cut paper strips simulating the fabric measuring and food dividing they do at home
- Observation prompt:** For Ms. Johnson: 'Note when students connect physical object to p/q notation — where does the bridge hold? Where does it break?'
- Identity opening:** Lesson opens: 'You already know how to do this. Today we learn to write what you already know.'

Generate community-informed lesson →

Before vs. After
Same topic. Same standard. Different lesson.

Generic fractions lesson	Opens with pan dulce — students' real experience
English only instruction	English + Spanish vocabulary side by side
Desk-based paper problems	Physical fold-and-cut embodied activity
No colleague connection	Observation prompt for Ms. Johnson embedded
Treats students as novices	Acknowledges their lived fraction knowledge

Figure 8. Prototype showing the redesigned form with community context saved and a side-by-side comparison of the current generic output versus the community-informed output.

3.2 Feature 2: Asynchronous Lesson Study Module

The Lesson Study module is a new platform section that creates genuine communities of practice across schools. Lesson Study — where teachers identify a shared challenge, design a lesson together, each teach it, observe student thinking, and reflect collectively — is among the most

research-supported professional development structures available (Lewis & Hurd, 2011). Most schools cannot implement it for lack of coordinated time. MagicSchool AI can make it happen asynchronously, across schools, at no cost. Teachers describe a learning challenge, are matched with four to six peers facing similar challenges, and work together over six to eight weeks: developing a shared research lesson with AI support, teaching it independently, submitting observation reports, reviewing AI-synthesized patterns across all classrooms, and documenting what they learned in a shared Community Knowledge Base accessible to all teachers on the platform. The AI facilitates at every step — it never decides. All substantive decisions belong to the teachers.

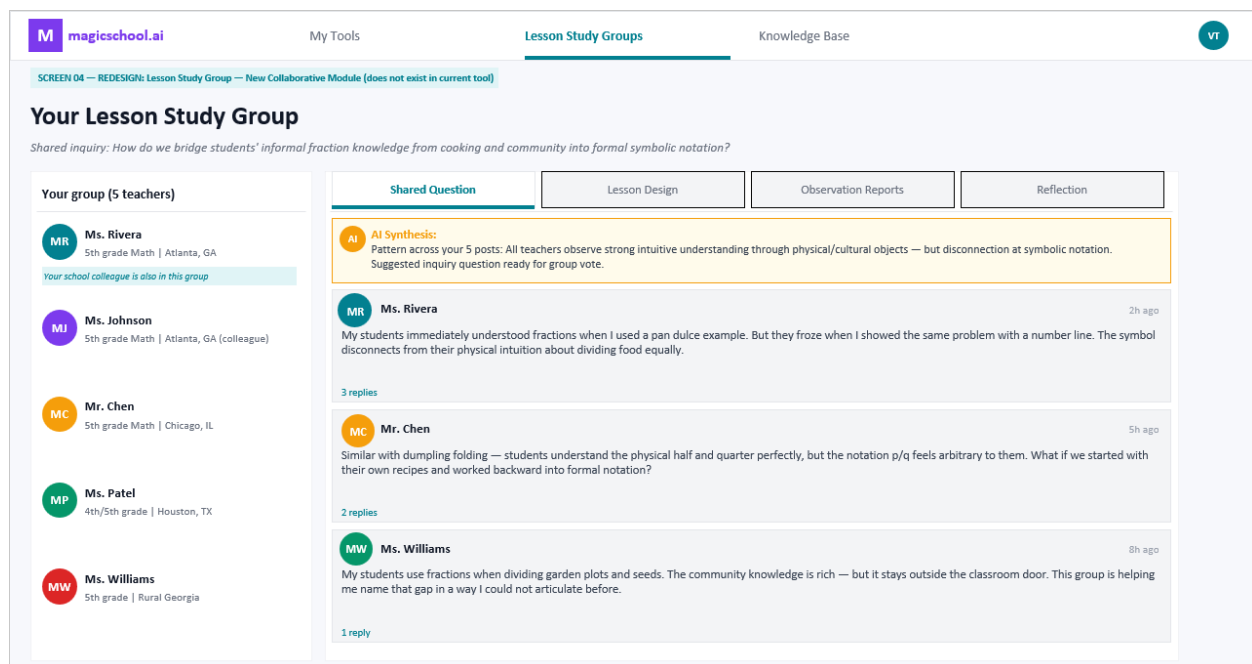


Figure 9. Prototype showing five teachers matched by shared inquiry, a collaborative workspace with AI synthesis, and teacher discussion posts building collective professional knowledge.

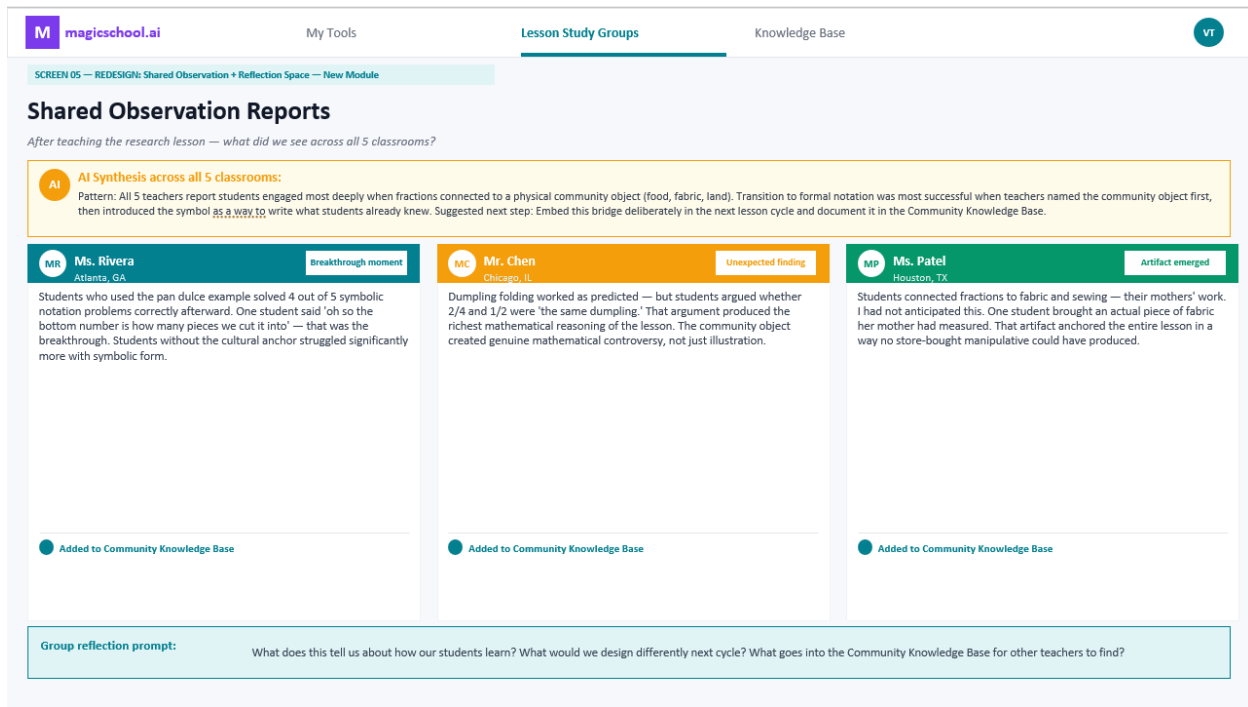


Figure 10. Prototype showing the post-teaching observation space with AI synthesis across five classrooms and individual observation reports archived in the Community Knowledge Base.

3.3 Current Design Versus Redesign

The comparison table below summarizes how the redesign changes MagicSchool AI across all eight sociocultural dimensions examined.

Dimension	Current MagicSchool AI	Redesigned
Participation	One teacher, one AI – no colleagues	Cross-school Lesson Study groups
Identity	Teacher = solo content producer	Teacher = community practitioner
Funds of Knowledge	Student background is never asked	8-question intake activates community knowledge

Situated Context	Generic decontextualized outputs	Student/school context shapes outputs
Embodiment	Text-based only	Observation protocols + embodied activities
Power Relations	AI decides what good teaching looks like	Teachers co-construct; AI facilitates
Third Space	Does not exist in the current design	Community intake creates this bridge
Assessment	Hours saved and completion rate	Participation growth and identity shift

Limitations of the Redesign

While the redesign significantly strengthens the sociocultural dimensions of MagicSchool AI, it also introduces practical constraints. Sustained participation in Lesson Study groups requires teacher time, coordination, and institutional support — resources that vary widely across schools. The Community Context Intake depends on teachers’ willingness to provide detailed information, which may be uneven across schools. Additionally, AI-generated synthesis must be transparent to avoid reproducing dominant cultural assumptions or flattening the nuance of teacher observations. These limitations do not undermine the redesign but highlight areas where thoughtful implementation and ongoing refinement are necessary.

Part 4: Assessing the Redesign

4.1 Evidence of Success and Assessment Methods

Assessing this redesign from a sociocultural perspective means asking different questions than completion rates and hours saved. Those metrics fit the acquisition metaphor — they tell us nothing about whether teachers are learning together, growing professionally, or becoming more responsive to their communities. The question that matters is: Are teachers participating more fully over time? Are they noticing students thinking more specifically? Are they drawing on community knowledge in their planning? Are they developing a different professional identity? Answering these requires methods from sociocultural research traditions — interaction analysis, discourse analysis, longitudinal participation tracing, and identity documentation.

The table below lays out five dimensions of assessment, the appropriate method for each, and what evidence of genuine learning would look like.

What we assess	Method	Evidence of success
Participation growth	Interaction analysis	Peer turns increase; quieter voices become active contributors
Student thinking	Discourse analysis of reflections	Teachers name specific student ideas, not general behavior
Community knowledge	Coding for funds of knowledge	Plans cite family and community context alongside standards
Identity shift	Longitudinal participation tracing	Movement from observer to full contributor over the semester
Transfer to design	Compare lesson plans pre/post	Final plans include cultural anchors and embodied activities

4.2 What a Successful Learning Experience Looks Like

A fifth-grade teacher at a Title I school in Atlanta completes the Community Context Intake, describes her Latinx bilingual students' embodied fraction knowledge as it emerges through daily cooking, and is matched with four teachers across Georgia, Illinois, and Texas who face the same challenge. Over six weeks, they design a shared research lesson, each teaches it, and submit observation reports. The AI synthesis reveals a consistent pattern: in every classroom, students who engaged with community-grounded examples demonstrated stronger symbolic reasoning, and several brought physical objects from home to anchor the abstraction. At the end of that cycle, this teacher has not simply produced a better lesson plan. She has worked through a genuine professional question with colleagues she would never have otherwise met, documented student thinking with specificity, and contributed situated knowledge to a shared base that other teachers can access. Her lesson plans look different — not because the AI improved, but because she now sees her students and her professional community differently. That is what the participation metaphor looks like when it is built into a platform by design. And it is what this redesign makes possible for five million teachers who currently use MagicSchool AI entirely on their own.

References

Gutiérrez, K. D. (2008). Developing a sociocritical literacy in the third space. *Reading Research Quarterly*, 43(2), 148–164.

Gutiérrez, K. D., & Jurow, A. S. (2016). Social design experiments: Toward equity by design. *Journal of the Learning Sciences*, 25(4), 565–598.

Jordan, B., & Henderson, A. (1995). Interaction analysis: Foundations and practice. *Journal of the Learning Sciences*, 4(1), 39–103.

Lave, J., & Wenger, E. (1991). *Situated learning: Legitimate peripheral participation*. Cambridge University Press.

Lewis, C., & Hurd, J. (2011). *Lesson study step by step: How teacher learning communities improve instruction*. Heinemann.

MagicSchool AI. (n.d.). MagicSchool for educators. <https://www.magicschool.ai>

Moll, L. C., Amanti, C., Neff, D., & González, N. (1992). Funds of knowledge for teaching: Using a qualitative approach to connect homes and classrooms. *Theory Into Practice*, 31(2), 132–141.

Sfard, A. (1998). On two metaphors for learning and the dangers of choosing just one. *Educational Researcher*, 27(2), 4–13.

Wenger, E. (1998). *Communities of practice: Learning, meaning, and identity*. Cambridge University Press.

